

Mini Vibration Motor Module

Hardware Overview

What is it?

The **Vibration Module** is a tiny DC motor with an unbalanced weight attached to its shaft. When the motor spins, the uneven weight causes the entire module to violently shake. It is exactly the same hardware used inside video game controllers and smartphones for haptic feedback.

Electrical Explanation

The Breakout Board (The Shield): When the motor it spins, it draws large amounts of current, and when it stops, it shoots a spike of reverse-voltage. If you plugged a motor directly into the microcontroller, it would instantly fry it. There is a transistor on the breakout board. It acts as an electronic floodgate, allowing the K-Comp to safely control the current for the motor.

PWM (Pulse Width Modulation): To control the intensity of the vibration, we do not lower the voltage. Instead, we turn the 5V power ON and OFF thousands of times per second. By changing how wide those ON pulses are, we control the speed of the motor.

Key Hardware Facts:

- **Connection:** Wired to a Digital PWM Pin (e.g. D6).
- **Voltage:** Operates at 5V.
- **Signal Flow:** K-Comp Processor → PWM Control Signal (Digital Pin) → Breakout Board (Transistor) → "High"-Current Power → Vibration Motor.

Software & Programming

kcomp_VibrationMotor.h

This library allows you to set a constant vibration speed or trigger short, non-blocking haptic "clicks" without freezing the rest of your program.

Usage: `#include <kcomp_VibrationMotor.h>`

Function Reference

Function Name	Description & Parameters
<code>void initVibration(uint8_t pin)</code>	Setup: Initializes the motor on a the PWM pin specified.
<code>void setVibration(uint8_t intensity)</code>	Action: Turns the motor on at a specific speed. ▪ <code>intensity</code>: 0 to 255.
<code>void stopVibration()</code>	Action: Instantly cuts power to the motor.
<code>void vibratePulseNonBlocking(uint8_t intensity, uint16_t durationMs)</code>	Action: Starts a temporary haptic pulse. ▪ <code>intensity</code>: 0 to 255. ▪ <code>durationMs</code>: Time in ms (e.g., 200 for a short click).
<code>void pollVibration()</code>	The Worker: If you use the pulse function, you MUST call this constantly inside <code>loop()</code> . It checks the clock and turns the motor off when the time is up.

⚠ Pitfalls & Troubleshooting

The "Stall" Voltage (It hums, but doesn't shake)

If you use `setVibration(20)`, the motor won't spin. This is because the magnetic force is too weak to overcome the physical friction of the unbalanced weight.

The Fix: You must supply an intensity of at least 80 to 100 to get the weight moving.

The Never-Ending Vibration

If you call `vibratePulseNonBlocking(255, 500)` but the motor turns on and never stops, you have forgotten the worker function.

The Fix: Ensure `pollVibration()` is written inside your `loop()`. If K-Comp isn't constantly checking the clock, it will never know when to shut the motor off.



Sample Program

This program demonstrates constant control and non-blocking haptic pulses via the Serial Monitor.

```
#include <kcomp_VibrationMotor.h>

#define VIB_PIN 6 // Must be a PWM capable pin

void setup() {
    Serial.begin(115200);
    initVibration(VIB_PIN);

    Serial.println("Send '1' for a short click.");
    Serial.println("Send '2' for a long, heavy rumble.");
    Serial.println("Send '3' to turn ON permanently at 50% power.");
    Serial.println("Send '0' to stop all vibration.");
}

void loop() {
    // CRITICAL: Must be called in loop IF you use vibratePulseNonBlocking!
    pollVibration();

    if (Serial.available() > 0) {
        char cmd = Serial.read();
        while (Serial.available()) Serial.read(); // Clear the serial buffer

        if (cmd == '1') {
            Serial.println("Short Click (Max Power, 100ms)");
            vibratePulseNonBlocking(255, 100);
        }
        else if (cmd == '2') {
            Serial.println("Heavy Rumble (Power 150, 800ms)");
            vibratePulseNonBlocking(150, 800);
        }
        else if (cmd == '3') {
            Serial.println("Permanent ON (Power 127)");
            setVibration(127); // 127 is exactly half of 255
        }
        else if (cmd == '0') {
            Serial.println("Emergency Stop.");
            stopVibration();
        }
    }
}
```

💡 **Try it out:** You can find the full sample program **SimpleVibrationMotorModule** in the K-Comp examples folder!